

Ashutosh Kumar Singh

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GitHub — LinkedIn

HCI-focused Embedded & Robotics Engineer with strong hands-on experience in **microcontroller-based prototyping, wearable/robotic systems, sensing, and real-time control**. Patent holder and published researcher with proven ability to **design–build–test physical systems**, integrate hardware with software, and document results rigorously.

Education

B.Tech — Information and Communication Technology (VLSI & Embedded Systems) 2023–2027

Marwadi University, Rajkot

CGPA: 8.62/10

Technical Skills

Programming & Firmware: C++, Python, Embedded C, Firmware Development

Microcontrollers & Platforms: Arduino, ESP32, NodeMCU, ARM Cortex

Sensors & Actuators: Load Cells, MQ2/MQ6 Gas Sensors, IMU, Motors

Communication Protocols: UART, SPI, I2C

Robotics & Control: ROS (Nodes, Topics), Sensor Fusion, Real-Time Control, PID

Wearable & HCI Systems: Physical Interaction, Wearable Devices, Human-in-the-Loop Testing

Design & Prototyping: KiCad (PCB Design), Proteus, Rapid Prototyping, 3D Printing

Tools: Git, Keil, Debuggers, Technical Documentation

Patents & Publications

Patent Granted — LPG Cylinder Monitoring & Leak Alert System

Patent No: **202421033238**

- Designed a safety-critical embedded system integrating load cells and gas sensors.
- Implemented real-time sensing, calibration, and alert mechanisms for household safety.

Journal Publication — AI-Based Resource Management for 5G Network Slicing

Concurrency and Computation: Practice and Experience (Wiley)

DOI: 10.1002/cpe.8327

- Co-authored a comprehensive review on AI-driven resource management for 5G/6G network slicing.

Projects

Autonomous Robotics Platform (ROS + Embedded Systems) 2025

- Developed a modular ROS-based robotic architecture with real-time sensor fusion and motor control.
- Integrated perception and control pipelines for autonomous navigation.

Hardware Trojan Detection & Analysis 2025

- Studied hardware Trojan insertion at RTL and gate level in digital circuits.
- Analyzed detection techniques using functional testing and side-channel parameters.

Work Experience

Embedded Systems Intern — Indian Army (Signal Corps) Aug 2025

- Validated critical communication hardware used in field operations.

Summer Intern — eInfochips (Hybrid) Jun 2025

- Developed firmware, serial communication logic, and conducted system integration testing.

Achievements & Leadership

Winner — National Hackathons (Hack4India, BrainRot, others) **4.5L Total Prize**

All India Rank **719 (97%ile)** — AME CET 2023

Convenor — Circuitology Club, Marwadi University

State-Level Chess Player